

The foundation laid out by previous generations

Experts call Windows 8 more secure than its previous versions but this didn't come about over just a couple of years. Windows 8 stands on a solid foundation of security laid out in Windows Vista, Windows 7 and XP and that's where we'll begin.

NTFS / EFS

NTFS, which provides enhanced security for files, is one of the most impressive file systems around. It comes with a sub-system called EFS (Encrypting File System) that can be used to encrypt files and folders. To encrypt the file (or folder), you open the file's properties and click on the 'Advanced' button on the bottom of the 'General' tab. In the 'Advanced Attributes' window, you can enable the 'Encrypt contents to secure data' option and apply it.

The best part about EFS is that it doesn't need any explicit decryption once you've encrypted the file. The user who has encrypted the file can open it just like any other normal file and EFS will encrypt/decrypt the file in the background as the user works on it. EFS is so tightly integrated into Windows that if the disk is taken out the system or the OS removed/reinstalled, it locks out the files completely. This behavior is embedded in the process of encrypting the files; here's how it happens:

1. The file is encrypted using either AES or 3DES algorithm with a randomly generated number as the FEK (File Encryption Key). The reason AES is used is because it's fast and symmetric. Once encrypted, the file can only be decrypted by using the FEK.
2. The FEK is encrypted using one of the most famous and secure public key algorithms – RSA. EFS saves the encrypted version of the FEK with the file. The public-private key pair to be used by RSA is automatically generated for a user if it's the first time s/he is encrypting a file. Since the FEK can be decrypted only by the private key, knowing the private key is necessary. This key is stored in the user's profile directory. At this stage, if the file is transferred to another computer it's protected because the FEK is encrypted and the private key needed to decrypt it won't be available.
3. Storing the private key so openly exposes a major security risk. Hence the private key is also encrypted by using a master key. At this point, the file can only be decrypted if the master key is available.
4. The master key is also generated randomly by Windows when the encryption happens the first time. The user's master key is encrypted using 3DES algorithm which employs a key that's partly based on the